

Midterm project

Project Plan for Capstone 1: System Analysis and design

Topic: Online Shopping System (User Registration and Purchasing)

Team name: AJOUTYATA

Team members:

Ibragimov Aziz 202590180 (Team Leader)

Artyom Kazakov 202590188 (Researcher)

Ibragim Muhamedjanov 202590206 (Diagram designer)

Tyo Vladislav 202590240 (Analysist)

Sxay Nikita 202590231 (Researcher)

Asalya Raimberdiyeva 202590221 (Ppt maker)

Online Shopping System (User Registration and Purchasing)

1. Project Plan

Project Objective

Develop an online shopping system that allows customers to register, browse products, add products to a shopping cart, place orders, and make payments.

Project Scope

The system will cover:

- User registration and login
- Product browsing
- Shopping cart
- Order placement
- Payment
- Basic product and order management

Target Users

- **Customer:** Registers, logs in, browses products, manages the cart, and purchases products.
- **Administrator:** Manages products and orders.

2. Use Case Modeling

Required: More than 3 Use Cases

Planned Use Cases

1. Register Account
2. Login
3. Browse Products
4. Manage Shopping Cart
5. Place Order
6. Make Payment
7. Manage Products
8. Manage Orders

Diagram Explanation

The **Use Case Diagram** will show the main actors and their interactions with the Online Shopping System. The Customer will interact with registration, shopping, ordering, and payment functions, while the Administrator will manage products and orders.

Use Case Specifications

Each use case will have a short specification describing:

- Actor

- Precondition
 - Main flow
 - Postcondition
-

3. Business Process Model (BPM)

Required: More than 2 BPMs

BPM 1 — User Registration Process

Shows the workflow from opening registration to entering information, validating the information, creating the account, and receiving confirmation.

BPM 2 — Product Purchasing Process

Shows the workflow from browsing products to adding products to the cart, placing an order, making payment, and receiving order confirmation.

BPM 3 — Order Management Process

Shows how the Administrator views an order and updates its status.

4. Chen ERD

Required: More than 3 Chen ERDs

Chen ERD 1 — User Registration

Shows the relationship between **User** and **Account**, focusing on user registration and account information.

Chen ERD 2 — Shopping Cart and Product

Shows the relationship between **User**, **Shopping Cart**, and **Product**, focusing on products selected by a customer.

Chen ERD 3 — Order and Payment

Shows the relationship between **User**, **Order**, and **Payment**, focusing on purchasing and payment information.

Chen ERD 4 — Order and Product

Shows the relationship between **Order**, **Order Item**, and **Product**, focusing on which products are included in an order.

5. IE ERD

Required: More than 3 IE ERDs

IE ERD 1 — User and Account

Shows the database relationship between the **User** and **Account** tables using primary and foreign keys.

IE ERD 2 — User, Shopping Cart, and Product

Shows how users, shopping carts, cart items, and products are connected in the database.

IE ERD 3 — User, Order, and Payment

Shows how a user creates an order and how the payment information is connected to that order.

IE ERD 4 — Order, Order Item, and Product

Shows how orders contain order items and how each order item is connected to a product.

6. Schedule:

Stage	Possible Time	What Will Be Done	Reason
1. Project Plan	Week 1	Define objectives, scope, and target users	This should be done first because it establishes what the system will include and who will use it.
2. Use Case Modeling	Week 1–2	Create Use Case Diagram and Use Case Specifications	Use cases identify the system's main functions and user interactions before creating process and database models.
3. BPM	Week 2–3	Create the 3 Business Process Models	BPMs should follow the use cases because they describe how

the identified activities happen step by step.

Chen ERDs can be developed after understanding the system processes because the processes help identify the required entities and relationships.

IE ERDs can be created after the Chen ERDs because they provide a more structured representation of the same database relationships.

The final review ensures that all required deliverables are complete and consistent with each

4. Chen ERD Week 3–4 Create more than 3 Chen ERDs

5. IE ERD Week 4–5 Create more than 3 IE ERDs

6. Review and Finalization Week 5 Check all diagrams, specifications, relationships,

and other before
requirements submission.

Overall Schedule: Approximately 5 weeks.

Reason for the Order:

The project starts with the **Project Plan**, then moves to **Use Cases, Business Processes, Chen ERDs**, and finally **IE ERDs**. Each stage provides information needed for the next stage, making the development process more organized and reducing inconsistencies between the diagrams.